

MESSSTETTEN, GermanyWMO: **108270**Lat: **48.1694N**Lon: **8.9433E**Elev: **900**StdP: **90.97**Time Zone: **1.00 (EUC)**Period: **94-12**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.2	-10.1	-14.3	1.2	-8.4	-12.1	1.5	-7.6	10.4	4.3	9.3	3.4	2.4	70	0.559

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.1	26.8	17.3	25.0	16.6	23.3	16.1	18.3	24.4	17.5	23.4	16.7	22.1	2.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.1	12.8	20.5	15.2	12.1	19.6	14.6	11.6	18.9	55.6	24.4	52.8	23.5	50.2	22.2	22.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.8	6.5	5.5	-14.8	29.5	2.5	1.6	-16.6	30.7	-18.0	31.6	-19.4	32.5	-21.2	33.7		
			-14.9	19.9	2.5	1.0	-16.7	20.7	-18.2	21.3	-19.6	21.9	-21.4	22.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.4	-1.1	-0.2	3.0	6.8	11.3	14.4	15.9	16.0	12.0	8.2	3.0	-0.6
	DBStd	7.54	4.49	4.96	4.62	4.45	4.03	4.02	3.58	3.64	3.73	4.21	4.45	4.42
	HDD10.0	1660	345	287	219	116	33	8	1	2	22	84	213	329
	HDD18.3	4033	603	520	476	347	221	128	93	91	193	313	461	587
	CDD10.0	718	0	0	2	19	72	139	185	188	81	29	3	0
	CDD18.3	49	0	0	0	0	2	9	18	18	2	0	0	0
	CDH23.3	369	0	0	0	1	13	72	126	149	9	0	0	0
	CDH26.7	58	0	0	0	0	0	7	19	32	0	0	0	0
Wind	WSAvg	2.6	2.7	2.9	3.0	2.7	2.6	2.4	2.4	2.2	2.4	2.5	2.5	2.7
Precipitation	PrecAvg	898	57	55	58	64	100	99	105	85	68	71	68	69
	PrecMax	1134	139	138	151	123	150	169	212	153	127	152	130	154
	PrecMin	671	4	18	15	8	42	46	52	17	26	11	3	1
	PrecStd	109	34	29	33	30	35	31	34	31	27	34	32	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	12.9	16.8	20.9	25.2	27.8	28.8	30.0	24.5	20.9	15.4	10.2
		MCWB	4.8	6.0	8.9	11.4	15.1	17.4	17.8	17.5	15.8	14.4	9.4	6.4
	2%	DB	8.0	10.2	14.0	18.7	22.7	25.5	26.5	26.9	22.0	18.2	12.6	7.8
		MCWB	5.2	5.1	7.5	10.3	14.4	16.8	17.4	17.4	15.1	13.2	8.5	5.2
	5%	DB	6.5	8.0	11.7	16.4	20.5	23.6	24.5	24.8	19.9	16.0	10.6	6.5
		MCWB	4.3	4.2	6.6	9.4	13.1	16.3	17.0	16.6	14.2	11.9	7.7	4.7
	10%	DB	4.9	6.1	9.5	14.2	18.4	21.7	22.5	22.6	17.9	14.0	8.9	5.2
		MCWB	3.0	3.5	5.9	8.5	12.2	15.5	16.1	16.0	13.3	11.1	6.8	3.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.4	7.2	9.8	12.2	16.7	19.2	19.3	19.3	17.5	15.2	10.5	7.4
		MCDB	8.3	10.9	14.4	19.0	22.3	24.5	25.4	25.9	21.9	19.3	13.4	8.8
	2%	WB	5.8	5.8	8.4	10.9	15.2	17.7	18.3	18.2	16.2	13.7	9.1	6.0
		MCDB	7.3	8.7	12.0	17.3	20.9	23.4	24.3	24.7	20.2	16.6	11.6	7.2
	5%	WB	4.5	4.8	7.3	9.8	13.8	16.7	17.5	17.2	15.0	12.6	8.1	4.9
		MCDB	6.3	7.5	10.5	15.3	19.3	22.2	23.3	23.3	18.6	15.1	10.2	6.2
	10%	WB	3.1	3.6	6.2	8.8	12.8	15.8	16.6	16.3	13.9	11.4	7.0	3.9
		MCDB	4.6	5.7	9.1	13.6	17.4	20.9	21.8	21.9	17.2	13.7	8.6	5.0
Mean Daily Temperature Range	5% DB	MDBR	5.6	6.3	7.2	8.4	8.9	9.1	9.1	9.0	8.1	7.5	5.8	5.1
		MCDBR	7.7	9.2	10.3	11.4	12.0	11.8	11.9	12.3	11.0	10.4	8.7	7.0
		MCWBR	5.4	5.7	5.9	5.8	6.1	5.9	5.5	5.8	5.7	5.9	5.5	5.2
	5% WB	MCDBR	6.8	7.9	8.9	10.5	10.8	10.8	10.8	11.1	9.8	9.3	7.6	6.0
		MCWBR	5.3	5.3	5.4	5.6	5.9	5.6	5.3	5.7	5.4	5.6	5.1	5.0
Clear-Sky Solar Irradiance	taub	0.279	0.301	0.332	0.363	0.373	0.377	0.377	0.373	0.350	0.327	0.299	0.276	
	taud	2.478	2.423	2.390	2.332	2.332	2.347	2.335	2.367	2.416	2.474	2.505	2.478	
	Ebn at Noon	811	863	886	888	889	884	878	867	855	820	776	767	
	Edh at Noon	62	82	99	116	121	120	120	111	96	77	61	55	
All-Sky Solar Radiation	RadAvg	1.03	1.81	3.01	4.26	4.99	5.72	5.41	4.73	3.52	2.17	1.17	0.83	
	RadStd	0.13	0.28	0.43	0.61	0.52	0.48	0.48	0.47	0.37	0.23	0.15	0.14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (103 neighbors)	N/A	N/A	N/A	N/A	+0.40	+0.37	N/A	N/A	+49	N/A